Graph Theory

Lecture 5

Advanced Structural Theory of Trees

Characterizations • Proofs • Degree Structure • Center • Robustness

What We Already Assume

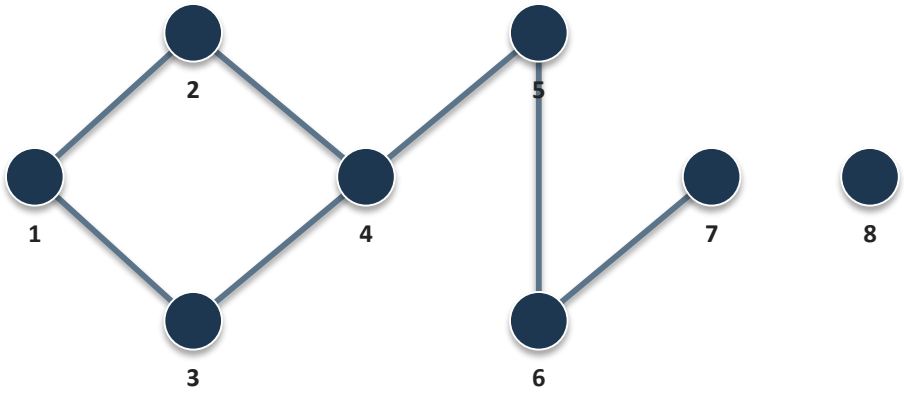
From Discrete Structures, students already know:

- tree = connected and acyclic graph
- leaves, internal vertices, rooted trees
- basic traversals and applications

Today we move from definitions to structural reasoning:

- How can a tree be recognized in several equivalent ways?
- What must its degree structure look like?
- Why is a tree simultaneously efficient and fragile?

Opening Problem: Is Edge Count Enough?



This graph has 8 vertices and 7 edges.

Is it necessarily a tree?

No — vertex 8 is isolated, while another component contains a cycle.

Conclusion: $|E| = |V| - 1$ alone is not sufficient.

What extra condition would make it sufficient?

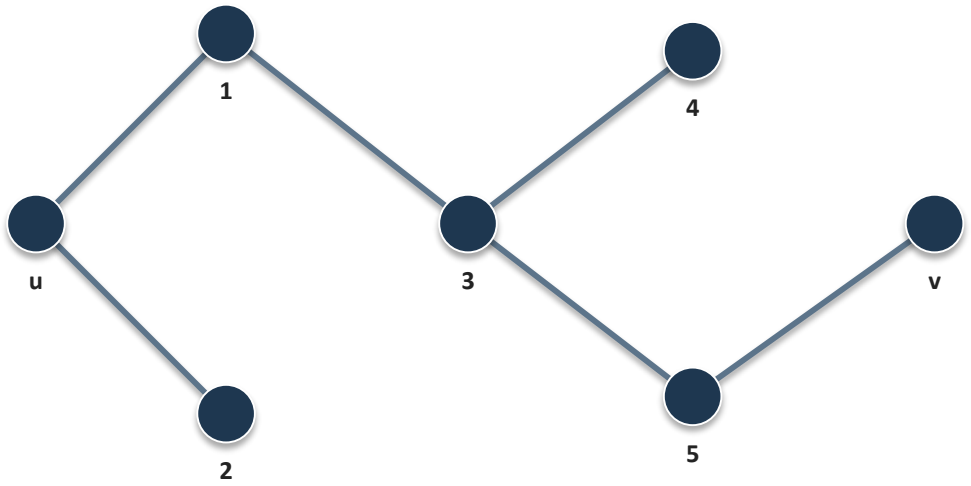
Equivalent Characterizations of a Tree

For a graph T with n vertices, the following viewpoints are equivalent:

- connected + acyclic
- exactly one path between every pair of vertices
- connected + n−1 edges
- acyclic + n−1 edges
- minimally connected
- maximally acyclic

These are not separate facts — they describe the same structure from different directions.

Characterization 1: Unique Paths



Pick any two vertices u and v.

In a tree there is exactly one u–v path.

Why is this stronger than merely saying the graph is connected?

Connectedness guarantees at least one path; tree structure guarantees exactly one.

Proof Idea: Tree ⇒ Unique Path

Assume T is a tree.

Because T is connected, a path exists between every pair u and v.

Suppose two distinct u–v paths existed.

Following one path from u to v and returning along the other creates a cycle.

But a tree is acyclic — contradiction.

Therefore every pair of vertices has exactly one path.

Proof technique: assume the forbidden structure exists, then derive a contradiction.

Reverse Direction: Unique Path ⇒ Tree

Assume every pair of vertices has exactly one path.

A path exists between every pair, so the graph is connected.

Suppose a cycle existed.

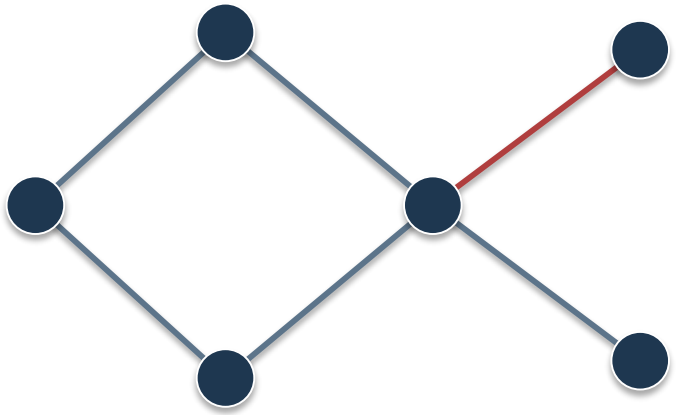
Choose two vertices on that cycle.

The two directions around the cycle give two distinct paths between them.

Contradiction.

Hence the graph is connected and acyclic — therefore it is a tree.

Characterization 2: Minimally Connected



A tree is connected, but it contains no unnecessary edge.

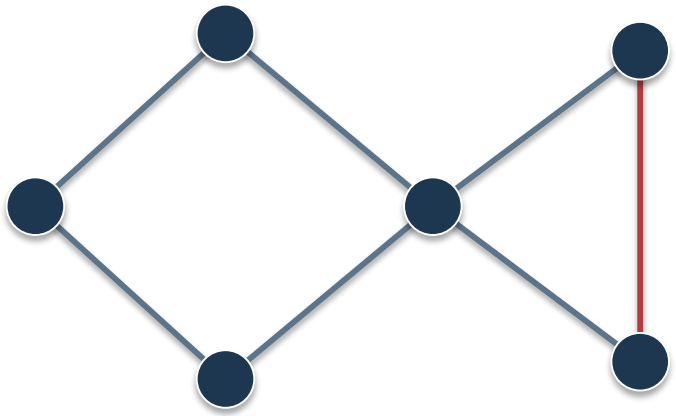
Remove any edge → the graph becomes disconnected.

Therefore every edge of a tree is a bridge.

This is why a tree is called minimally connected.

Connected with zero edge redundancy

Characterization 3: Maximally Acyclic



Start with a tree: no cycles exist.

Add any edge between two previously nonadjacent vertices.

The tree already contains a unique path between those endpoints.

The new edge closes that path into exactly one cycle.

So a tree is maximally acyclic.

Structural View: Trees Sit at a Boundary

Remove any existing edge	**TREE**	**Add any missing edge**
Disconnected	**Connected + Acyclic**	**Exactly one cycle**

This viewpoint is useful in network design: a tree uses exactly enough edges to stay connected — no more.

Why Does Every Tree Have n−1 Edges?

If $|V(T)| = n$, then $|E(T)| = n - 1$

We will prove this using induction.

Key structural fact needed first:

Every nontrivial tree has at least one leaf — in fact at least two.

Removing a leaf and its incident edge leaves a smaller tree.

Inductive Proof of $|E(T)| = n-1$

Base case: $n=1 \rightarrow 0$ edges, so $|E|=n-1$.

Inductive step:

- Choose a leaf v and remove it with its single incident edge.
- The remaining graph $T-v$ is a tree on $n-1$ vertices.
- By induction, $T-v$ has $n-2$ edges.
- Restore v and its one edge: $|E(T)|=(n-2)+1=n-1$.

Notice how the proof exploits a structural operation specific to trees: leaf deletion.

Degree Structure of a Tree

$$\Sigma \deg(v) = 2|E| = 2(n-1)$$

This follows by combining the Handshaking Lemma with $|E|=n-1$.

It imposes strong restrictions on possible degree sequences of trees.

Example question: Can a 7-vertex tree have degrees 4,3,2,2,1,1,1?

Check the sum before trying to draw the tree.

A Stronger Leaf-Count Relation

$$n_1 = 2 + \Sigma(i-2)n_i \quad \text{for } i \geq 3$$

n_i = number of vertices of degree i.

Interpretation:

high-degree branching vertices force additional leaves.

Immediate consequence:

every nontrivial tree has at least two leaves.

This relation lets us infer the number of leaves without drawing the tree.

Worked Example: Infer the Leaves

Suppose a tree has:

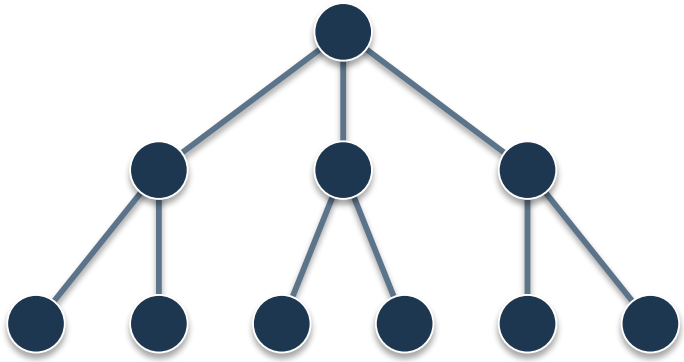
- 3 vertices of degree 3
- 1 vertex of degree 4
- all other non-leaf vertices have degree 2

Then:

$n_1 = 2 + 3(3-2) + 1(4-2)$

$n_1 = 2 + 3 + 2 = 7$

So the tree must have 7 leaves — regardless of its drawing.



Distance Structure: Eccentricity, Radius and Diameter

For a vertex v, eccentricity e(v) is the greatest distance from v to any vertex.

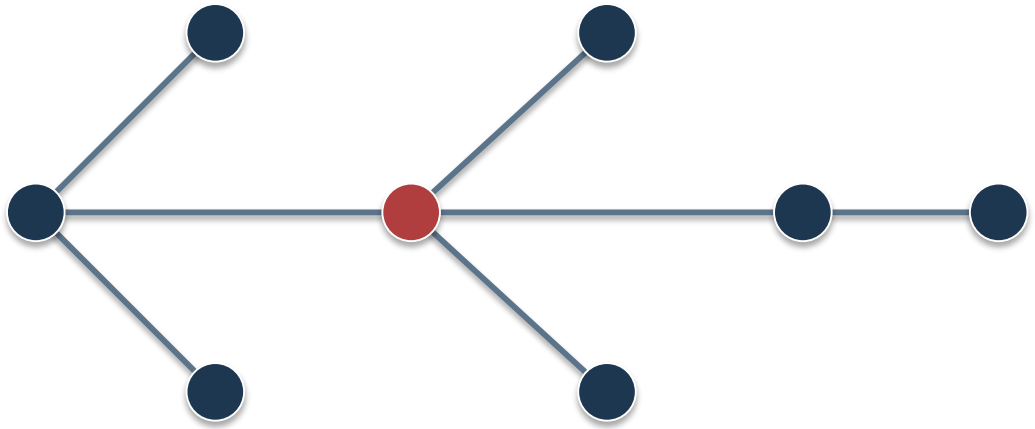
Radius = smallest eccentricity in the tree.

Diameter = largest eccentricity; equivalently, the length of a longest path.

Center = vertex/vertices having minimum eccentricity.

Important theorem: a tree has either one center or two adjacent centers.

Finding the Center by Repeated Leaf Removal



Algorithmic idea:

- remove all current leaves simultaneously;
- repeat on the remaining tree;
- stop when one vertex or two adjacent vertices remain.

The survivor(s) form the center.

Application: facility/server placement when minimizing worst-case distance.

Important Distinction: Center ≠ Highest-Degree Vertex

The center is determined by distances, not by degree.

A high-degree vertex can still be far from one end of the tree.

Therefore:

- degree measures local connectivity;
- eccentricity measures global distance position.

Graph measures answer different structural questions — do not substitute one for another.

Trees and Network Robustness

Every edge is a bridge → λ(T)=1

A nontrivial branching tree has articulation points

Engineering interpretation:

- Tree networks use the minimum number of links required for connectivity.
- But they have no alternate edge route.
- One critical link failure can disconnect the system.

Therefore tree design exposes a fundamental trade-off:

Minimum Cost ↔ Redundancy / Fault Tolerance

Lecture Summary

Trees admit several equivalent structural characterizations.

Unique paths explain both minimal connectivity and maximal acyclicity.

A tree with n vertices has exactly n−1 edges.

Degree-sum relations allow structural information such as leaf counts to be inferred.

Center, radius and diameter describe global distance structure.

Trees are edge-efficient but inherently weak against link failures.

Next lecture: spanning trees, multiplicity, MSTs and network optimization.